

ENS 491-492 – Graduation Project

Final Report

Project Title: Graph-First Fall Detection with a
Constrained Bandit Decision

Group Members: Ceren Duru Çınar, Hasan Kuşpınar,
İlhan Sertelli, Filiz İpek Oktay

Supervisor(s): Prof. Selim Saffet Balcısoy

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1. EXECUTIVE SUMMARY

Falls are a major cause of injury and loss of independence among older adults, yet existing fall-detection systems continue to suffer from high false-alarm rates and limited interpretability. Many current approaches rely solely on body posture estimation or wearable sensors, which often fail to distinguish hazardous falls from benign activities such as lying on a bed or sitting down abruptly [10].

This study introduces a graph-first fall detection framework in which each video frame is represented as a structured scene graph, followed by an adaptive decision process based on a constrained contextual bandit. In the perception stage, multiple state-of-the-art computer vision models are integrated, including GroundingDINO for object detection, DPT-Large for monocular depth estimation, YOLOv8-Pose for human pose estimation, Mask2Former for instance segmentation, and SegFormer for semantic segmentation [5],[20],[21],[22]. Floor geometry is estimated via a RANSAC-based plane fitting procedure applied to depth values within floor-labeled regions, while a support-scoring mechanism associates each detected person with the most probable supporting surface, such as the floor, bed, or chair [18]. These components enable the construction of scene graphs with explicit relational predicates, including `standing_on`, `lying_on`, and `falling_towards_floor`.

In the decision stage, temporal changes between consecutive scene graphs are summarized through a compact delta representation capturing posture transitions, new floor contacts, and support changes. This representation is provided as input to a constrained LinUCB contextual bandit, which balances detection sensitivity against operational constraints such as maximum alarm rate and enforced cooldown periods [4]. The system is evaluated on two public benchmark datasets, URFall and CAUCAFall, by aggregating frame-level bandit outputs into a video-level fall probability and reporting standard performance metrics including accuracy, precision, recall, and F1-score [6],[16]. The resulting pipeline demonstrates a proof-of-concept integration of structured scene understanding and adaptive, interpretable decision-making for fall detection.

2. PROBLEM STATEMENT

Unintentional falls among elderly individuals represent a significant public health concern due to their high incidence, associated healthcare costs, and potential for severe or fatal injuries. Automatic fall-detection systems aim to enable early intervention; however, many existing approaches continue to exhibit high false-positive rates for non-hazardous activities such as lying on a bed, bending, or sitting down, as well as false negatives in cluttered indoor environments with occlusions and varying viewpoints [6],[10]. These limitations reduce the reliability of fall-detection systems in safety-critical monitoring scenarios.

Current commercial and research-oriented solutions are predominantly based on wearable accelerometer sensors or convolutional neural network (CNN)–based vision models. Wearable devices depend heavily on user compliance and correct placement, and may be uncomfortable, forgotten, or intentionally removed, limiting their practicality in continuous monitoring applications [6]. Vision-based approaches alleviate some of these issues but are often driven by pose estimation or end-to-end CNN classifiers, which can misinterpret visually similar body configurations when contextual information is ignored. In particular, purely pose-based methods frequently fail to distinguish between lying on elevated furniture and lying on the floor, a distinction that is critical for reliable fall detection [6],[10].

Recent advances in deep learning have improved fall-detection performance through enhanced pose pipelines, three-dimensional CNN architectures, and real-time object detectors [1],[7]. Nevertheless, these models are typically black-box in nature and provide limited interpretability regarding their decisions, which is undesirable in medical and elderly-care applications where trust and transparency are essential [10]. Moreover, most existing methods do not explicitly model human–object support relationships or capture how posture and support surfaces evolve over time, resulting in brittle behavior under realistic indoor conditions.

Adaptive decision-making mechanisms, such as contextual bandits, have recently been explored in healthcare monitoring tasks, demonstrating the potential to balance detection performance with operational constraints [4]. However, their integration with structured, relational scene representations remains largely unexplored in fall detection. Existing work typically applies bandit-based reasoning to high-level activity predictions rather than to explicitly modeled geometric and relational information derived from the scene [4].

This project addresses the problem of reliable and interpretable fall detection by combining structured scene understanding with adaptive, constraint-aware decision-making. The objective is to investigate whether explicit modeling of floor-plane geometry, human posture, and support relationships—encoded as scene graphs—and their temporal evolution can reduce false alarms while maintaining high sensitivity to true fall events. By integrating these representations with a constrained contextual bandit framework based on Constrained LinUCB, the proposed approach aims to provide adaptive yet regulated alarm decisions that are suitable for real-world, safety-critical deployment.

2.1. Objectives/Tasks

2.1.1. Graph-first perception pipeline

The first objective is to construct a perception pipeline that integrates multiple vision models into a unified framework. GroundingDINO, DPT-Large, YOLOv8-Pose, Mask2Former, and SegFormer are combined to produce pixel-level depth maps, instance masks, and human pose estimates for each frame[5],[20],[21],[22].

2.1.2. Floor-plane estimation

To support reliable reasoning about contact with the ground, the floor plane is estimated using the RANSAC algorithm on depth values corresponding to pixels classified as floor[18]. The intended result is a parametric plane model that enables robust computation of a person's distance to the floor.

2.1.3. Human posture classification

Human posture classes including standing, sitting, lying, and falling_towards are defined using joint angles and torso inclination derived from pose estimation. This task aims to produce consistent per-person posture labels for every frame.

2.1.4. Support score formulation and reasoning

A support score function is designed by combining mask underlap and overlap, intersection-over-union, horizontal and vertical alignment, depth differences, and mask quality. This enables identification of the most likely support object for each person, such as the floor or a piece of furniture, and allows encoding of relations like standing_on, sitting_on, lying_on, and falling_towards_floor.

2.1.5. Scene graph construction

Using the outputs of posture classification and support reasoning, scene graphs are constructed for each frame. These graphs encode human-object support relations and spatial relations such as left_of, right_of, above, below, in_front_of, and behind, and are stored as JSON triples in the form of subject, relation, and object.

2.1.6. Bandit-based temporal decision layer

A temporal decision layer is implemented using a constrained LinUCB contextual bandit. A 13-dimensional feature vector is extracted from consecutive scene graphs to summarize posture drops, new floor contacts, support changes, and other meaningful transitions, enabling adaptive and interpretable alarm decisions.

2.1.7. Alarm explanation and interpretability

For each alarm decision, a human-readable explanation is generated based on the underlying scene graph changes. This improves transparency, facilitates debugging, and supports validation of the system's behavior.

2.1.8. Evaluation on public datasets

The final objective is to evaluate the proposed system on the URFall and CAUCAFall datasets[6],[16]. Scene graphs and bandit decisions are generated for all videos, and performance is assessed at the video level using accuracy, precision, recall, and F1-score metrics.

2.2. Realistic Constraints

2.2.1. Computational constraints

The perception stage relies on multiple large deep learning models, including transformer-based detectors and segmentation networks, making per-frame processing computationally expensive. Real-time or edge deployment is therefore challenging. To address this constraint, the project focuses on offline video processing and uses careful batching and resolution selection instead of real-time optimization.

2.2.2. Data availability constraints

The evaluation is based on public fall datasets, namely URFall and CAUCAFall, which contain controlled scenarios[6],[16]. These datasets are relatively small and may not capture the full diversity of real-world environments, camera viewpoints, and body types. As a result, the findings are framed as a proof-of-concept rather than a complete clinical validation.

2.2.3. Annotation and ground-truth limitations

Ground-truth labels in the datasets are typically provided at the video level as fall or no fall, while detailed frame-level annotations for posture, support relations, or scene graphs are unavailable. The system therefore operates under a weak supervision setting, inferring fine-grained relationships without explicit frame-level labels and evaluating performance primarily at the video level.

2.2.4. Privacy and deployment constraints

Camera-based fall detection systems raise significant privacy concerns in clinical or home environments. Real deployments would require anonymization, secure data handling, and potentially on-device processing. Due to these concerns, the project is limited to offline analysis using public, anonymized datasets.

2.2.5. Standards and safety considerations

Although the project is conducted as a research study and is not subject to medical device regulations, any real-world deployment would need to comply with healthcare standards and data protection regulations. The constrained bandit design anticipates such requirements by explicitly limiting alarm frequency and incorporating cost-aware decision-making to reduce alarm fatigue and improve safety.

3. METHODOLOGY

The proposed system is organized into three main stages: perception and scene graph construction, delta-based temporal analysis using a constrained contextual bandit, and video-level fall scoring and evaluation. This modular structure allows complex visual information to be progressively transformed into interpretable and decision-ready representations.

3.1. Perception and Floor Estimation

The first stage consists of a perception pipeline that fuses the outputs of multiple pretrained computer vision models in order to obtain a comprehensive understanding of the scene. GroundingDINO is employed for open-vocabulary object detection, providing bounding boxes and class labels for relevant objects such as persons, beds, chairs, and tables. Monocular depth estimation is performed using the DPT-Large model, which produces a dense depth map for each pixel. Human pose information is extracted using YOLOv8-Pose, yielding joint locations including shoulders, hips, knees, and ankles. Instance-level object masks are obtained via Mask2Former, while SegFormer is used for semantic segmentation to assign semantic labels, such as floor, to each pixel[5],[20],[21],[22].

Using the output of the semantic segmentation model, pixels classified as floor are identified and paired with their corresponding depth values from the depth map. A planar surface model of the form

$$Z = AX + BY + C$$

is then estimated using the RANSAC algorithm, where X and Y denote normalized pixel coordinates and Z represents the estimated depth[18]. The use of RANSAC ensures robustness against outliers arising from misclassified or occluded floor pixels. The resulting plane model enables computation of the distance between any point on a person's mask and the floor surface, which is a critical factor for distinguishing between lying on the floor and lying on elevated surfaces such as beds or sofas.

3.2. Posture and Support Reasoning

Human posture estimation is derived from the pose keypoints obtained via YOLOv8-Pose [22]. Two primary geometric descriptors are computed: the torso inclination angle, defined as the angle between the vector connecting the hip midpoint to the shoulder midpoint and the vertical axis, and the knee angles for both legs, calculated from the hip–knee–ankle joint triplets. These features enable a coarse yet robust posture classification into four discrete states: standing, sitting, lying, and falling_towards. Standing is characterized by a small torso inclination and large knee angles, sitting by moderate knee flexion with relatively upright posture, lying by a torso orientation close to horizontal, and falling_towards by intermediate-to-large torso inclination combined with atypical joint configurations indicating a transition toward the ground.

To determine which surface supports each person, a support score is computed between every detected person and each candidate support object, including both the floor and furniture items. The score integrates several geometric and spatial cues and is defined as

$$S = q_m \cdot (\alpha U + \beta O + \gamma IoU + \delta H + A + D)$$

where

- U denotes the underlap ratio, representing the proportion of the person's mask located above the support mask,
- O represents the overlap ratio between person and support masks,
- IoU is the intersection-over-union of their bounding boxes,
- H measures horizontal alignment between the two objects,
- A captures vertical alignment between the bottom of the person and the top of the support surface,
- D denotes the difference between the median depth values of the person and support masks, and
- q_m is a mask quality factor that assigns higher weight to instance-level masks and lower weight to box-derived masks.

The weighting coefficients are chosen such that contact-related terms dominate purely spatial alignment terms, satisfying $\alpha > \beta > \gamma > \delta$. For each person, the support object with the highest score is selected, and a posture-dependent relation is assigned.

Specifically, a standing person supported by the floor is labeled as standing_on, a sitting person supported by furniture as sitting_on, a lying person supported by either the floor or a bed as lying_on, and a person in the falling_towards posture whose strongest support candidate is the floor is assigned the relation falling_towards_floor.

3.3. Scene Graph Construction

Each video frame is subsequently represented as a scene graph in which nodes correspond to detected objects and persons, and edges encode relationships between them. Spatial relations such as *left_of*, *right_of*, *above*, *below*, *in_front_of*, and *behind* are derived from bounding box centers and depth overlap. In addition, human-specific support and posture relations, including *standing_on*, *sitting_on*, *lying_on*, and *falling_towards_floor*, are explicitly encoded. The resulting scene graphs are serialized as JSON-formatted triples of the form (subject, relation, object), with human-readable identifiers such as *person_1*, *bed_1*, and *floor_1* preserved to facilitate debugging, inspection, and explanation. An example frame from the URFall dataset and its corresponding scene graph representation are illustrated in Figures 1 and 2, respectively.



Figure 1. An example frame from the URFall dataset.

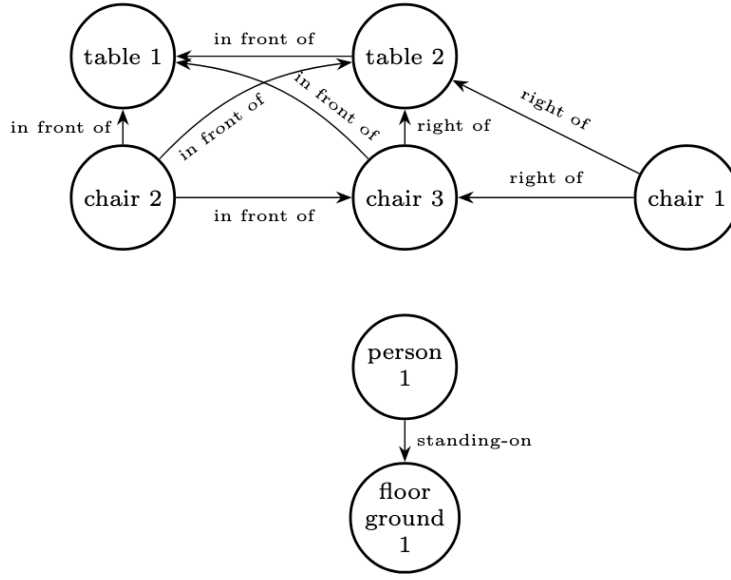


Figure 2. An image representing the scene graph of the frame in Figure 1.

3.4. Delta Summary and Constrained LinUCB

To capture temporal dynamics, scene graphs from consecutive frames are compared to produce a delta summary that quantifies meaningful changes over time. These changes include posture drops such as transitions from standing or sitting to lying, the emergence of new floor contacts, support changes such as transitions from bed to floor, the appearance or disappearance of persons, significant reversals in spatial relations, and the first occurrence of falling_towards_floor relations.

From this delta summary, a 13-dimensional feature vector is constructed to serve as the contextual input to a constrained LinUCB bandit. This vector includes the overall delta score, computed as a weighted combination emphasizing posture drops and floor-related events, as well as counts of posture transitions, floor contacts, and support changes, normalized change rates per person, and binary indicators signaling the presence of any person touching the floor or exhibiting a falling_towards_floor relation.

The Constrained LinUCB agent operates with two possible actions: raising an alarm or withholding an alarm. For each action, the bandit maintains separate parameter estimates and computes an upper confidence bound that balances expected reward and uncertainty. Action selection is predominantly greedy with respect to the upper confidence bound, while a small exploration probability is retained to ensure continued adaptation.

Operational constraints are incorporated into the decision process to reflect real-world deployment requirements. These include a K-in-N rule requiring the delta score to exceed a threshold in at least K of the last N frames, a maximum allowable alarm rate within a fixed time window, a cooldown period following each alarm, and a cumulative cost budget limiting excessive alerts. When an alarm is generated, it is accompanied by a concise textual explanation derived directly from the delta summary and scene graph changes, for example indicating posture transitions, support changes, and the presence of falling_towards_floor events. This explanation mechanism enhances transparency and interpretability.

The bandit is updated online using a reward signal determined by the eventual classification of alarms as true positives, false positives, false negatives, or true negatives. Correct detections slightly reduce the decision threshold, false alarms increase it, and missed falls lead to a more aggressive reduction. Through this adaptive thresholding mechanism, the system gradually tunes its sensitivity to the characteristics of the data and the surrounding noise, enabling robust and context-aware fall detection over time.

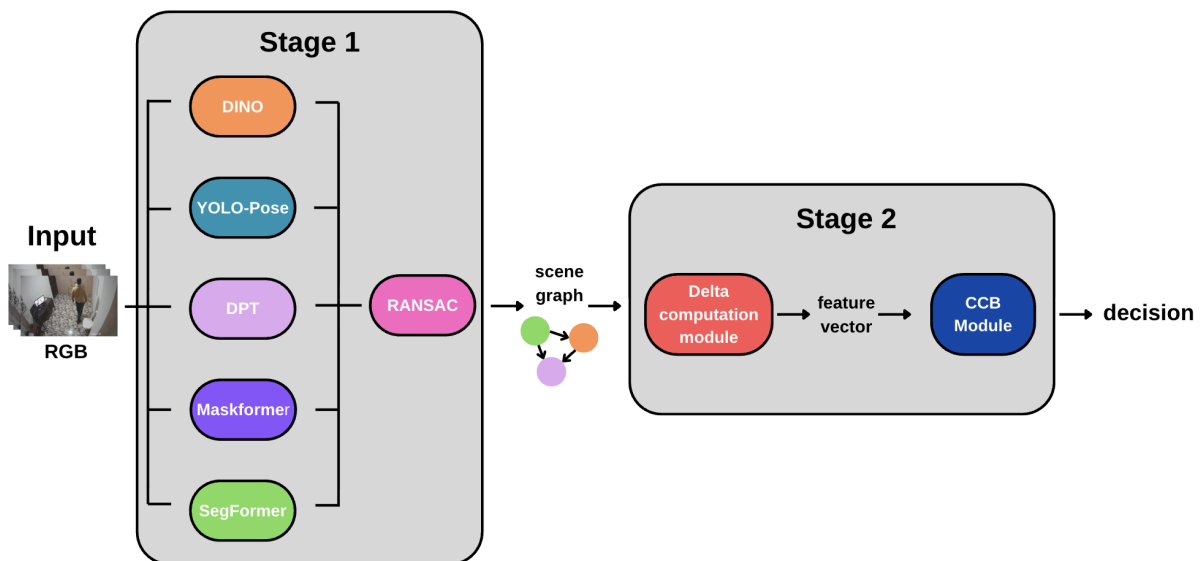


Figure 3. The diagram of the pipeline.

3.5. Video-Level Fall Scoring and Evaluation

When the constrained bandit produces one or more alarm decisions within a video, these frame-level outputs are aggregated to compute a single video-level fall probability. Rather than relying on isolated alarms, the system summarizes the temporal behavior of posture, support, and floor-contact events across all sampled frames, providing a more stable and interpretable final decision.

For each video, several ratios are computed by analyzing the bandit decisions and the underlying scene graph transitions. These include the proportion of frames exhibiting posture drops, the proportion of frames with newly observed floor contact, the proportion of frames involving support changes related to the floor, the proportion of frames in which the person remains in a *lying_on floor* state, and the average depth deviation of the person from the RANSAC-estimated floor plane during floor-contact periods. Together, these quantities capture both the occurrence and persistence of fall-related structural changes over time.

These ratios are combined with manually selected weights into a raw score s_{raw} as

$$s_{raw} = 2.5 r_{post} + w_f r_{floor} + w_s r_{sup} + 2.0(r_{dur} + \beta_{dur} + 1.0 d_{avg} + b)$$

- r_{post} denotes the fraction of frames containing posture drops,
- r_{floor} denotes the fraction of frames with newly observed floor contact,
- r_{sup} denotes the fraction of frames involving floor-related support changes,
- r_{dur} denotes the proportion of frames in which the person remains lying on the floor,
- d_{avg} denotes the average depth deviation from the estimated floor plane,
- w_f is a weighting coefficient for floor-related components
- β balances the contribution of floor-contact duration, and
- b is a bias term.

The raw score s_{raw} is mapped to the interval $[0, 1]$ using a sigmoid function,

$$S = \frac{1}{1 + e^{-s_{raw}}}$$

where S represents the estimated probability that the video contains a fall event. Higher values of S correspond to a higher likelihood of a fall. Based on empirically determined thresholds, videos are classified into three categories: *fall present*, *suspicious*, or *no fall*.

Finally, the predicted video-level labels are compared against the ground-truth annotations provided by the datasets. System performance is evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score, allowing quantitative assessment of both fall detection capability and false alarm behavior.

4. RESULTS & DISCUSSION

4.1. Quantitative Results

The quantitative performance of the proposed fall-detection system was evaluated using video-level annotations from the URFall and CAUCAFall datasets [6],[16]. Since both datasets provide ground-truth labels at the video level, system outputs were aggregated temporally and evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score.

The overall classification behavior of the system is summarized by the confusion matrix shown in Figure 4. This matrix illustrates the distribution of true positives, true negatives, false positives, and false negatives across the evaluated videos. The results indicate that the system successfully identifies a substantial portion of fall events while maintaining a controlled number of false alarms, reflecting the effect of the constrained decision-making mechanism.

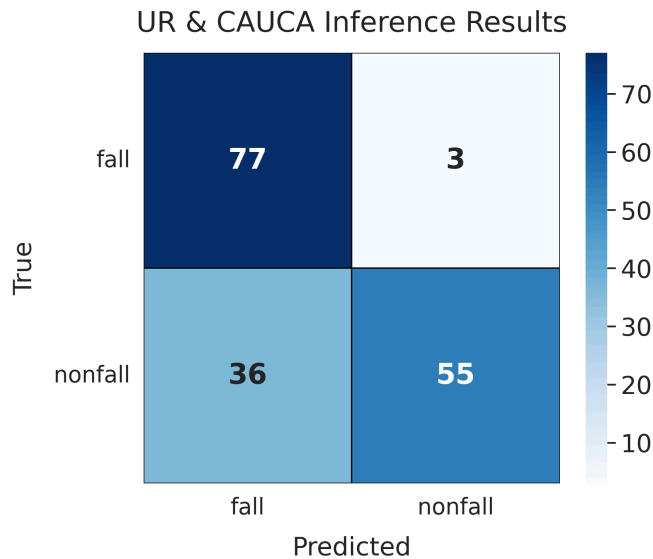


Figure 4: Confusion Matrix of the Inferences

A more detailed quantitative comparison is provided in Table 1, which reports precision, recall, F1-score, and accuracy metrics for the proposed method alongside several multimodal and vision-language baselines, including InstructBLIP, SmolVLM2-500M, Video-LLaVA-7B, and Qwen3-VL-8B-Instruct [2],[3],[19],[23]. Despite not relying on end-to-end training or large-scale multimodal pretraining, the proposed approach achieves competitive performance across all reported metrics. While larger models such as Video-LLaVA-7B and Qwen3-VL-8B-Instruct benefit from extensive pretraining and achieve high overall accuracy, they often exhibit trade-offs between recall and false positives.

Method	Precision	Recall	F1-score	Accuracy
InstructBLIP (Baseline)	0.471	1.000	0.640	0.471
SmolVLM2-500M	0.824	0.735	0.710	0.735
Video-LLaVA-7B	0.784	0.783	0.783	0.782
Qwen3-VL-8B-Instruct	0.929	0.924	0.924	0.924
Our Method (LinUCB)	0.686	0.960	0.800	0.760

Table 1: Quantitative Representations of Performances Across Different Models

In contrast, the proposed pipeline is designed to provide a more balanced performance profile, maintaining a high recall value while avoiding excessive false alarms. High recall is particularly important in emergency scenarios such as fall detection, where missing a true fall event can have serious consequences. By explicitly modeling posture, floor contact, and contextual relationships, the system is able to reliably capture fall events while preserving a stable F1-score, demonstrating that structured, context-aware reasoning can be an effective and practical alternative for safety-critical applications.

To complement these aggregate metrics, qualitative frame-level examples are included to illustrate representative outcomes contributing to the confusion matrix. In Figure 5, the frame showing a person lying on the floor corresponds to a true positive detection, where the system correctly identifies a fall event based on posture transition, floor contact, and sustained support by the floor. This case contributes to the true positive count and demonstrates successful integration of posture reasoning and floor-plane estimation. In contrast, the frame in Figure 6, depicting a person bending forward and picking an object off the ground, represents a false positive case, highlighting one of the main sources of confusion in the pipeline.

Although no actual fall occurs, the posture configuration and transient body inclination partially resemble fall-like dynamics, leading the system to overestimate fall risk. This example contributes to the false positive entries in the confusion matrix and illustrates a known limitation of posture-driven reasoning when contextual cues are ambiguous or short-lived.



Figure 5: True Positive Example (Fall Event)



Figure 6: False Positive Example (Bending Toward Floor)

4.2 Realization of the Initial Objectives

The initial objective of this project was to design and implement a vision-based fall detection pipeline by leveraging multiple state-of-the-art computer vision models to extract meaningful information from video data. This objective was largely achieved through the successful integration of object detection, pose estimation, depth estimation, and semantic and instance segmentation models into a unified perception framework. These components enabled the system to reason about human posture, spatial layout, and interactions with the surrounding environment, forming the basis of a functional fall-detection pipeline.

In particular, the project accomplished its goal of moving beyond posture-only fall detection by incorporating contextual cues such as floor geometry and human–object support relationships. By explicitly modeling whether a person is supported by the floor or by furniture, the system was able to distinguish between visually similar activities that commonly lead to false alarms in conventional approaches. Overall, the implementation fulfills the initial project objectives by delivering a complete computer-vision–driven fall detection pipeline that can analyze realistic indoor scenes and identify fall-related events in video sequences.

4.3 Deviations from the Initial Objectives

Despite the successful realization of the core goals, several deviations from the initial project plan occurred. Originally, the system was intended to rely more on large language models operating directly on video data. However, early experiments revealed that such approaches were computationally expensive, difficult to constrain, and less suitable for safety-critical decision making. As a result, the final design shifted toward a structured perception pipeline combined with a constrained contextual bandit. This deviation ultimately strengthened the system’s interpretability and controllability. Instead of black-box LLM-based decisions, alarms are now triggered based on explicit scene graph transitions and rule-based summaries by a constrained-contextual bandit module. The benefit of this shift in the approach is evident in Figure 7, where each alarm is accompanied by a concise, human-readable explanation derived from observable changes in posture and support relations rather than implicit model decisions.

```
{
  "significant_change": true,
  "confidence": 0.8655084520194857,
  "reason": "linucb: p1 floor contact; score=3.402
    .00",
  "score": 3.4,
  "delta_text": "Frame 1 summary:\n- p1: support
    floor\n- p1 to the right of table2\n- context
    p1: posture=lying_on, support=floor1, on_floor
    =True",
  "dynamic_threshold": 2.1899999999999996,
  "frame": 140
}
```

Figure 7. An example decision output by constrained-contextual bandit

4.4 Project Completion Status

The project can be considered successfully completed within the defined scope. A complete fall-detection pipeline was implemented, covering spatial perception, human posture analysis, contextual reasoning, and video-level evaluation. All major system components were integrated into a coherent workflow capable of processing video sequences and producing fall-related decisions, supported by both quantitative metrics and qualitative examples.

Although the system is not optimized for real-time deployment and relies on offline processing, these limitations were anticipated given the computational cost of the employed models. Within the defined scope of the study, the implementation demonstrates technical completeness, functional correctness, and a clear alignment between the initial problem definition, the system design, and the obtained results.

4.5 Contribution to State-of-the-Art

This project contributes to existing fall-detection research by demonstrating the feasibility of a context-aware, computer-vision-based pipeline that explicitly models scene structure rather than relying solely on posture or end-to-end classification. By integrating multiple perception models and reasoning about floor geometry and human-object support relationships, the system addresses known weaknesses of conventional vision-based approaches, particularly in distinguishing falls from visually similar non-fall activities.

While the proposed solution does not outperform existing state-of-the-art methods in terms of raw accuracy, it presents a structured and interpretable alternative to

traditional fall detection approaches. By combining geometric reasoning with contextual information from the scene, the system demonstrates how fall detection decisions can be made more understandable and transparent. The proposed pipeline serves as a proof-of-concept and shows a clear direction for future improvements and extensions.

5. IMPACT

This project has scientific, technological, and socio-economic impacts in the domain of intelligent healthcare monitoring, with a focus on fall detection for elderly care environments.

5.1 Scientific Impact

This project's primary scientific contribution is a graph-first approach to fall detection. The suggested system explicitly models scene context by representing each frame as a scene graph encoding posture, support surfaces, and spatial relations, in contrast to conventional approaches that mainly rely on posture or wearable sensors. This makes it possible to distinguish between visually similar scenarios—like lying on a bed versus lying on the floor—that are frequently the cause of false alarms with greater reliability.

The project also shows how a constrained contextual bandit can be used for temporal decision-making in fall detection. The system maintains interpretability while bridging the gap between static classifiers and realistic safety-critical monitoring systems by fusing structured scene representations with adaptive decision rules and operational constraints.

5.2 Technological Impact

In terms of technology, the project offers a modular pipeline that combines several cutting-edge computer vision models into a single framework for perception and decision-making. The system is more extensible and simpler to maintain thanks to the scene graph abstraction, which offers a clear interface between the perception and decision layers.

Since each alarm can be linked to specific posture changes or support transitions, the use of explicit scene relations and delta-based summaries enhances explainability and transparency. This is especially crucial for applications in the healthcare industry, where system accountability and operator trust are vital.

An additional illustration of how such a system can be incorporated into a multi-camera supervisory setting is depicted in Appendix A, B, C. The mentioned interface allows streaming from multiple cameras and highlights detected falls or possible falls directly on the screen. Rather than displaying raw model outputs, the interface provides visual assistance such as bounding boxes (red or yellow based on risk) derived from the pipeline. A moment where the system is alerting a fall and a possible fall can be seen in

Appendix C. This design allows for a better monitoring system which also involves humans. Including humans in the decision making process can be helpful in decreasing the effect of false positives.

5.3 Socio-Economic Impact

Falls are a leading cause of injury and hospitalization among older adults. The suggested system may speed up response times and lessen caregiver workload by using contextual reasoning and alarm constraints to minimize false alarms and missed detections. The system demonstrates how intelligent monitoring tools can assist caregivers without taking the place of human judgment, even though it was only tested on publicly available datasets.

5.4. Innovation, Commercial Potential, and FTU

When compared to other fall detection methods, the combination of scene-graph-based perception and constrained bandit decision-making represents a novel strategy. Extension to additional safety-critical situations and settings is made possible by the modular design.

Since the system depends on pretrained models with different open-source licenses, any commercial deployment would need additional real-world validation, real-time optimization, and a Freedom-to-Use (FTU) analysis.

6. ETHICAL ISSUES

The proposed pipeline relies on video-based monitoring of rooms in a care facility, which can raise ethical concerns related to privacy, data protection and bias. Even though the system is designed to support security or care personnel rather than to intervene, sensitive data of individuals still needs to be handled. Below, the key ethical issues and mitigation steps are listed.

6.1. Privacy Concerns

Camera based monitoring systems often raise concerns about the privacy of the subjects. In this project, the subjects are older adults and caregivers, and they express discomfort about being monitored continuously and fear misuse of data [7], [8], [17]. Videos, especially monitoring living areas may capture sensitive moments, making them prone to inappropriate access. In this project, these concerns are addressed by using only public anonymized datasets (URFall, CAUCAFall) with no real personal data, avoiding any storage of identifiable frames beyond temporary processing and using scene-graphs for later steps of the pipeline, where only relations (e.g., *person lying_on floor*) are transmitted instead of video frames. Future implementations can incorporate on-device inference and privacy-enhancing methods similar to those presented in [12], [14], [15].

6.2. Data Protection and Regulations

Under KVKK and GDPR, video data is part of highly sensitive personal data. The European Data Protection Board explicitly warns that video devices must follow principles of data minimization, necessity, and limited retention [9]. Although this project only uses public datasets and does not rely on real patient data, a deployed system would need to ensure no long-term storage of raw video unless explicitly justified, that videos are used solely for fall detection, not for worker monitoring or analytics and secure access control to prevent unauthorized viewings. In real deployments, organizations must complete Data Protection Impact Assessments (DPIA) as recommended by the EDPB [9].

6.3. Bias and Dataset Limitations

Many fall-detection datasets rely on young, healthy volunteers simulating falls, which do not accurately reflect the behaviors or physical characteristics of older adults [24]. This may lead to generalization biases, which means the system may perform differently based on differences in age and body type. In addition, in detection and pose estimation, computer vision systems are known to show demographic performance disparities, as demonstrated by Buolamwini and Gebru's study on algorithmic bias [13]. These disparities may lead to higher false-negative rates for underrepresented groups.

This project reduces reliance on direct body shape by using scene context, support relationships and floor plane geometry. Therefore, it decreases dependence on pose estimation alone. Nevertheless, the limitation is acknowledged and the system is classified as proof-of-concept rather than clinically validated. Future work should evaluate performance across diverse populations, mobility patterns, and environments.

6.4. Misclassification Risk and Human Safety

Incorrect classifications in fall detection can have severe consequences. False negatives may delay emergency response, risking severe injury [7],[24]. False positives may cause caregivers to not trust or ignore alerts over time. To mitigate misclassification risks, the system includes a CCB limiting alarm frequency and enforcing cooldown periods and rule-based verification (posture drops + new floor contacts) before issuing alarms.

6.5. Freedom to Use and Licensing

In this project, pretrained models like YOLO, SegFormer, Mask2Former, GroundingDINO, which all have varying open source licenses, are used for perception [5],[11],[21],[22]. While for academic research all are permissible, deploying this pipeline would require license compatibility checks to avoid FTU violations. Academic usage terms in this project are fully complied, but any commercialization would mean reviewing model licenses and potentially retraining models on unrestricted datasets.

7. PROJECT MANAGEMENT

This project has a timeline of 2 semesters. Before the first semester started, the project topic was proposed by the supervisor. The initial proposed idea was using multimodal LLMs with audio for understanding. The group started to experiment with sending audio recordings to LLMs. However, in later steps the project evolved to continue with video rather than video and audio. In the first semester, the idea was still to use LLMs with videos. After initial experiments with certain LLM models, it was realized that LLMs were heavy in terms of parameters and cost. Therefore a decision was made to continue with the model pipeline consisting of smaller models including an LLM with smaller parameters as the last layer. By this decision, the group continued to work on a pipeline using different models, and experimented with this pipeline.

In summer, the group continued to have weekly meetings with the supervisor. In these meetings the first step of the pipeline was changed from using only raw videos to constructing a scene graph to focus on relevant changes in the scene. After this decision, the group focused on constructing scene graphs with different models. However, when published models had dependency and version issues, the decision was made to extract them manually, rather than using models.

After extracting the scene graphs, the task was to focus on how the LLM can work with the scene graph and how it can understand it. For this, certain ways such as LoRA and training a transformer were tried, yet they were not successful. In a regular weekly meeting, by a suggestion of the supervisor a new algorithm named Constrained Contextual Bandit was tried instead of using an LLM in the last layer. Initial experiments with the algorithm were successful, so it was decided that CCB would be used as a decision making step.

Initially, the thought was to have a verifier LLM as a last layer to verify the decisions of CCB. However, as the project progressed, making this system as an assistive tool and leaving the verifying to the security or care personnel was preferred rather than a verifier LLM. That is why a security panel with 3 screens was prepared. In this security panel, if the pipeline classifies a video as a fall or fall risk, the event is indicated with a bounding box (red or yellow based on fall or fall risk), and the bounding box of the video lights up. This was the final plan of the project.

In terms of managing the project, there are several important issues learned. From the beginning, all group members knew that long term projects are prone to changes because every part of the plan might not work. The most important thing is to work together and consult the supervisor whenever necessary. Also, being open to constructive criticism and changes is another significant issue. When all the members work considering these, the project is managed well in spite of changes and some errors.

8. CONCLUSION AND FUTURE WORK

8.1 Conclusion

This project proposed and evaluated a graph-first fall detection pipeline that integrates adaptive decision-making with structured scene understanding. Each video frame was represented as a scene graph encoding posture, support relations, and spatial context by combining several perception models for object detection, pose estimation, depth estimation, and segmentation. In order to enable adaptive fall detection while enforcing useful alarm constraints, temporal changes between successive graphs were condensed into compact delta features and analyzed using a constrained contextual bandit.

The project's primary result is a proof-of-concept system that shows how ambiguity in fall detection can be decreased by explicitly modeling floor geometry and human-object relationships. Because alarms are directly connected to observable events like posture drops, new floor contacts, or support transitions, the suggested method offers better interpretability than posture-only reasoning. The system can differentiate fall scenarios from non-fall activities while controlling false alarms through constraint-aware decision-making, according to evaluation on public datasets (URFall and CAUCAFall) [6], [16].

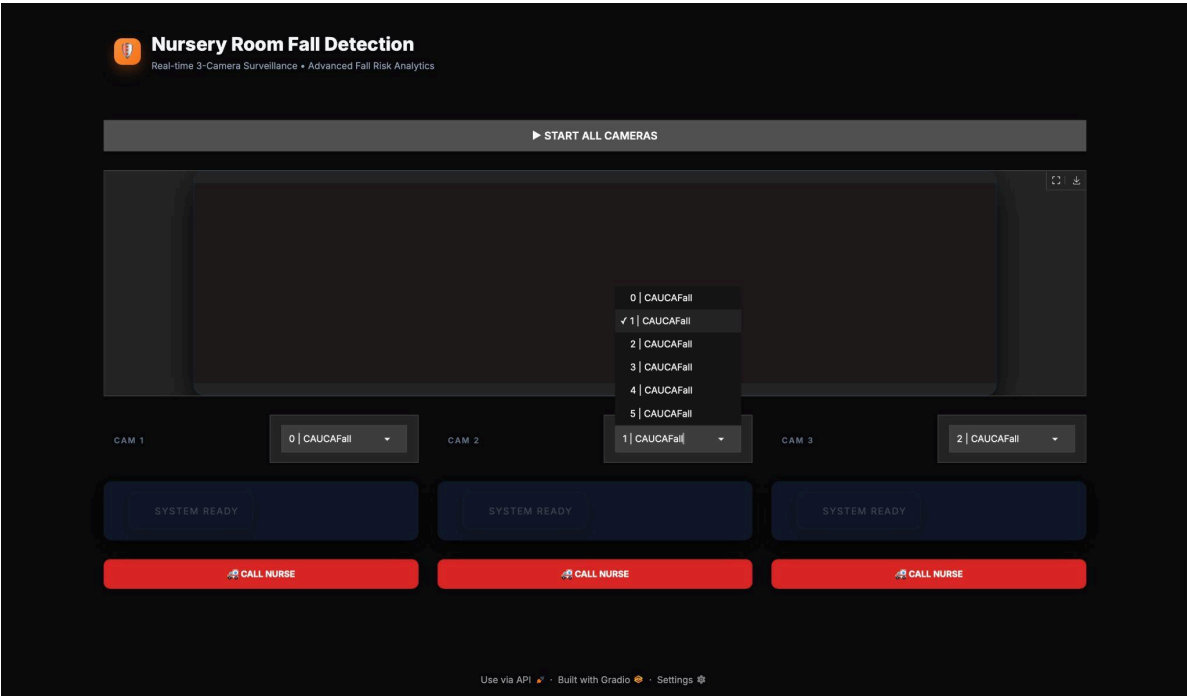
In spite of these advancements, a number of restrictions still exist. First, real-time deployment is difficult without additional optimization because the system depends on computationally costly perception models. Second, the assessment is restricted to publicly available datasets that simulate falls, which might not accurately represent variations in senior behavior, surroundings, or camera positioning in the real world. Third, fine-grained quantitative validation of posture and support reasoning is not possible because ground-truth annotations are only available at the video level. Therefore, rather than indicating clinical readiness, the conclusions should be read as proving feasibility.

8.2 Future Work

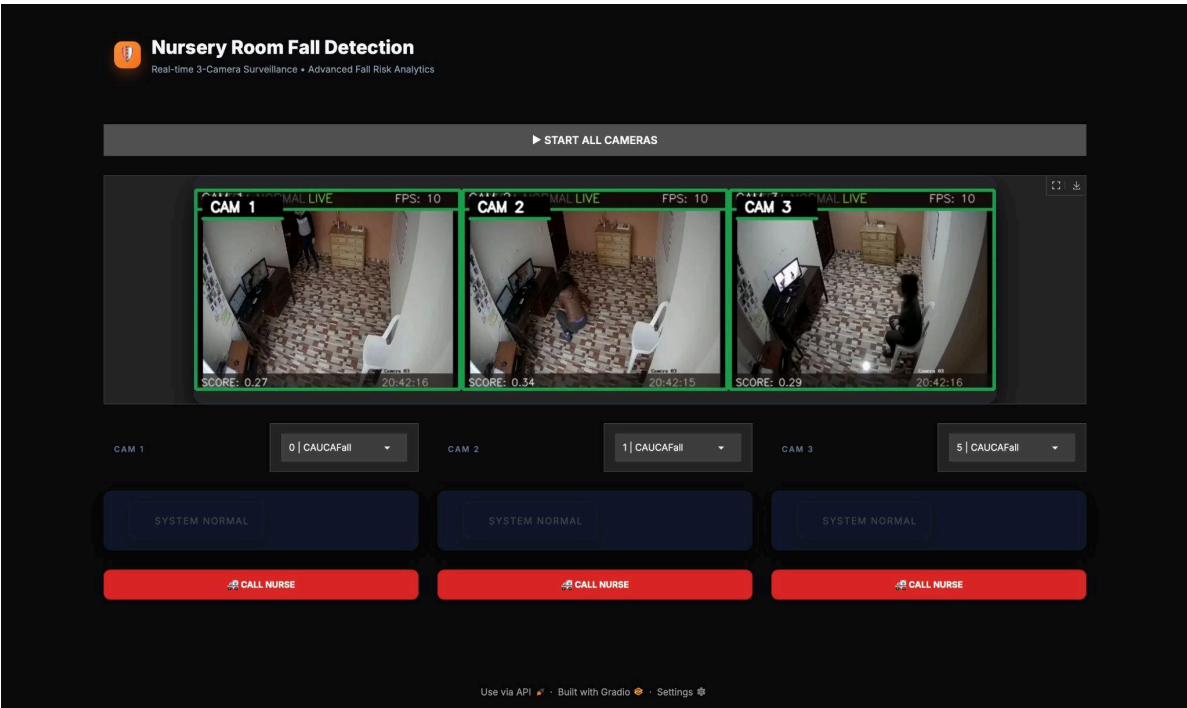
Future work can proceed in several directions. One possible direction for future work could include optimizing the pipeline for real-time or edge-computing purposes to eliminate redundancies in the models or utilize more perception-light models. Another direction could include optimizing the bandit framework for learning with temporal memory or adapting to individual environments. More real-world elderly data could be beneficial for analyzing the biases of the results.

Finally, there exists potential beyond the area of fall detection due to the abstraction using a scene graph. Since the general idea is to use a scene graph for person-object relations, the proposed framework can be extended to capture a wider range of risk-related situations without being limited to a single event definition. A context-aware and interpretable video monitoring system has been given a good beginning with this research work.

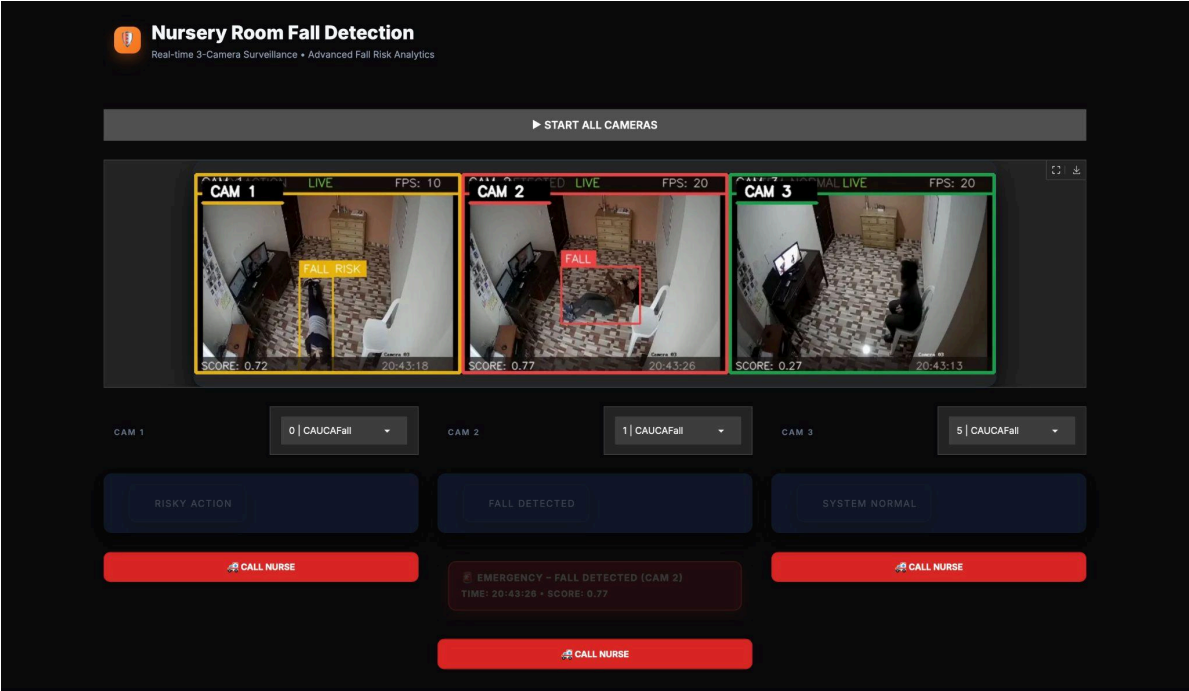
9. APPENDIX



Appendix A. Initial User Interface, 3 Camera Setup



Appendix B. Beginning of the Camera Processing



Appendix C. Alerted Emergency Fall Scenes

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